

Research paper

A Multi-step ahead photovoltaic power forecasting model based on TimeGAN, Soft DTW-based K-medoids clustering, and a CNN-GRU hybrid neural network



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ABSTRACT

Accurate photovoltaic (PV) power generation forecasting is very important for making economic and reliable power dispatching plans. This study proposes a multi-step ahead PV power forecasting (PPF) model, which combines time-series generative adversarial networks (TimeGAN), soft dynamic time warping (DTW)-based K-medoids clustering algorithms, and a hybrid neural network model computed by a convolutional neural network (CNN) and gated recurrent units (GRU). First, a new PV power data augmentation method based on TimeGAN is proposed to realize the expansion of the historical PV power data. Second, the combination of soft-DTW and K-medoids clustering is developed to extract and distinguish the commonalities and differences between time series characteristics of PV power. Finally, a hybrid neural network model is developed to integrate CNN and GRU into a unified framework for an accurate PPF under different weather conditions. The performance of the proposed model was tested using real datasets of two PV power stations with an installed capacity of 50MW. The comparative analysis of the proposed model with five reference models confirms its superiority in terms of forecasting accuracy. For the sunny days and cloudy days, the range of root mean square error (RMSE) values are produced by the proposed model distributed in [0.927–2.523] MW and [1.781–6.173] MW respectively with respect to the multi-step ahead forecasting from 15-min up to 24-hour. The experimental results of the hybrid model proposed in this study indicate that data augmentation-based clustering method is very helpful for improving the accuracy of PV power forecasting.

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1. Introduction

For the purpose of carbon reduction and carbon neutrality, the power system in transition is migrating into higher new renewable penetrations. Such as wind and PV generations. Jenkins et al. (2018). However, PV power generation presents obvious stochastic volatility characteristics. When a large number of PV installations are connected to the grid, it will cause a significant fluctuation in the grid voltage, current, and frequency and resulting in a lower power quality (Das et al., 2018). In order to eliminate above adverse effects, PPF as a tool has been proposed along with the economical dispatch. A reliable PPF is particularly

important to improve the stability of power system operation and PV power availability.

According to the forecasting time horizons, the PPF includes ultrashort-term forecasting, short-term forecasting, medium-term forecasting, and long-term forecast (Zang et al., 2020a), while the ultrashort-term and short-term PPF is the research object of this paper. For the PV systems, the solar irradiance received by PV arrays is usually affected by various meteorological factors, leading to the PV power fluctuation characteristics under different weather conditions (Shi et al., 2012). This means that the forecasting accuracy of the PV power depends not only on the selected forecasting pattern but also on the weather conditions. Therefore, to accurately forecast the uncertainty and fluctuation of PV output power caused by different weather patterns, the PPF models that are based on weather classification have been lured a great attention by the researchers (Das et al., 2017).

The key to improve the forecasting accuracy of the PV power relies on the extraction of training data, the establishment of the forecasting model, and the selection of forecasting methods. That

Abbreviations: TimeGAN, Time-series generative adversarial network; Soft DTW, Soft dynamic time warping; CNN, convolutional neural network; GRU, Gated recurrent units

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Nomenclature

ANN	Artificial Neural Network
CNN	Convolution Neural Network
CGAN	Conditional GAN
DTW	Dynamic Time Warping
DBA	DTW Barycenter Averaging
FeTGAN	Federated Time-Series GAN
GRU	Gated Recurrent Units
GAN	Generative Adversarial Networks
GASVM	Genetic algorithm-based Support Vector Machine
HKCG	(Soft DTW-based K-medoids)-(CNN-GRU)
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
PV	Photovoltaic
PPF	Photovoltaic power forecasting
SSE	Sum of squared errors
SIOA	Swarm intelligence optimization
TimeGAN	Time-Series Generative Adversarial Networks
RMSE	Root Mean Square Error
R^2	Determination Coefficient
RGAN	Recurrent GAN
RCCAN	Recurrent Conditional GAN

the selection of training samples closely related to the samples to be predicted is the basis for reliable outcomes. The forecasting methods with strong nonlinear learning ability can usually ensure high 1-step prediction accuracy. However, to achieve a higher multi-step forecasting accuracy, the establishment of good forecasting models is very important. In the PPF field, more research focuses on verifying the effectiveness of the relevant models through one-step forecasting. In fact, multi-step forecasting is even more challenging, and a good forecasting model should maintain good robustness within a certain range of forecasting steps (Qu et al., 2019). The existing PPF methods can be categorized into physical, statistical, and hybrid methods (Ma et al., 2021). A detailed comparison of previous studies in PPF is listed in Table 1.

In recent years, artificial intelligence algorithms based on statistical methods have been widely used in PPF (Liu et al., 2020; Dong et al., 2020). However, the PV power is affected by many meteorological factors, which make it difficult to obtain a high forecasting accuracy by a single method. Therefore, the hybrid forecasting model such as the combination of physical and statistical methods (Ogliari et al., 2017), swarm intelligence optimization (SIOA)-based ANN method (Seyedmahmoudian et al., 2018), signal decomposition–reconstruction methods (Li et al., 2021), and hybrid artificial neural network (ANN) based methods were proposed successively (AlShafeey and Csáki, 2021). The SIOA method can also find the optimal parameters of ANN, which can help the ANN method to achieve a better forecasting accuracy. But the SIOA-based method has poor training efficiency for large scale of datasets (Zang et al., 2020a).

The CNN-LSTM (Long short-term memory) based hybrid neural network model has been widely used in various time-series forecasting (Alhussein et al., 2020; Bogaerts et al., 2020; Qu et al., 2021) and has become the most popular regression-based forecasting method. Wang et al. (2019) used the hybrid CNN-LSTM model for day-ahead PV power forecasting, and the statistical results show that the hybrid model has higher accuracy than the single model (LSTM and CNN). Sajjad et al. (2020) present a

hybrid CNN-GRU model for electricity forecasting, which presents the smallest error compared to CNN, LSTM and CNNLSTM models. This combination benefits from the strong feature extraction ability of CNN and the strong temporal feature learning ability of LSTM and GRU, a higher forecasting accuracy can usually be obtained.

A similar weather condition would have a similar PV power generation (Liu et al., 2019). At present, the clustering technology based on unsupervised learning has been widely used for the data modeling and analysis of PV power station (Matterti et al., 2021). Li et al. (2018) proposed an envelope-based clustering technique, where the original PV power data is clustered into 4 categories in order to assist to build the more accurate forecast models. Due to the flexibility in sequence-alignment, the DTW method shows obvious advantages concerning time series dissimilarity measures, as compared with the traditional Euclidean distance method (Müller, 2007; Caillaud et al., 2020). A previous study (Zhakiya, 2018) suggests that DTW-based K-means can find anomalous events in continuous seismic time series. In Ref. Chen et al. (2017), a DTW distance-based K-medoids method was proposed to divide the buildings with similar social media activities into functional areas. DTW-based data analysis may not be sufficient due to the loss of an adequate averaging technique. The DTW Barycenter Averaging (DBA) clustering algorithm is a highly efficient technique for solving problems encountered with the existing pairwise averaging methods (Petitjean et al., 2011). However, the DTW method provides only local but not global optima. Cuturi and Blondel (2017) proposed to include a smoothing DTW algorithm, namely, soft-DTW, which is differentiable in all of its parameters. The experimental results presented in Ref. Cuturi and Blondel (2017) indicate that the soft-DTW algorithm is superior to DBA in computing barycenters and clusters from time-series data.

PV power stations are usually built in dry areas with abundant light resources, where there are more sunny and cloudy days but fewer rainy days. At present, the forecasting accuracy of the PV power generation on sunny days meets the requirements of the power dispatching departments due to the regularity in the PV power generation curve (Akhter et al., 2019). However, on cloudy and rainy days, due to the direct impact of cloud movement and obstruction to solar irradiance, the power curve has no stable transformation law. The forecasting accuracy for the overall power generation is low, resulting in penalties for the PV power generation companies (Zhang et al., 2020). Especially on rainy days, the PV power is not only random, but also the number of samples is small, which poses great challenges for effective PPF.

Data-driven models often require significant quantities of training data to ensure generalization. Data augmentation strategy is considered to be a method that can effectively improve the forecasting and classification accuracy when the amount of data is limited (Lu et al., 2021; Salamon and Bello, 2017). In Ref. Nie et al. (2021), the augmentation method based on a fixed number of sample points was used to enrich the sky image datasets for cloudy conditions, which reduced the short-term PPF error (RMSE) by an average of 4%. GAN-based data enhancement methods have become one of the most recent research topics (Antoniu et al., 2017; Kaur et al., 2020). GAN can directly generate new samples based on historical data, and the probability distributions of the data do not have to be specified explicitly. In addition, GAN uses unsupervised learning, which can avoid the cumbersome manual labeling (Chen et al., 2018). Esteban et al. (2017) propose RGAN (Recurrent GAN) and RCCAN (Recurrent Conditional GAN) to generate realistic real-valued time series. Based on RCCAN, TimeGAN, as proposed in Yoon et al. (2019), combines the flexibility of the unsupervised GAN framework with the control afforded by supervised training

Table 1

A detailed comparison of PPF methods with previous research work.

Ref.	Year	Method	Region	Forecasting horizon	Research direction	Lowest metrics
De Giorgi et al. (2016)	2016	GLSSVM (GMDH-LSSVM)	Italy	1 h, 3 h, 6 h, 12 h, 24 h ahead	Forecasting the PV output power at several time horizons up to 24 h	NMAE = 2.92%
Behera et al. (2018)	2018	PSO-ELM	India	15 min, 30 min, 60 min ahead	Solar photovoltaic power forecasting at several time horizons	MAE = (0.0144–0.0419) (p.u) RMSE = (0.0178–0.0534) (p.u)
VanDeventer et al. (2019)	2019	GASVM	Australia, Malaysia	1 h ahead	Short-term power forecasting of residential scale PV system	RMSE = 11.226 (W)
Eseye et al. (2018)	2018	Wavelet-(PSO-SVM)	China	3 h, 6 h, 12 h, 24 h ahead	Short-term PV solar power forecasting based on SCADA and meteorological information	MAPE = 4.22%
Khatib et al. (2022)	2022	LSTM and GRU	Palestine	1 h ahead	One-step ahead photovoltaic output current prediction	MAPE = (8.9–19.2)%
Zhang et al. (2021)	2021	Kmeans-improved BP	China	1 h ahead	One-step ahead photovoltaic power output prediction	RMSE = 0.11 (MW)
Massaoudi et al. (2021)	2021	NARX-LSTM	Qatar, Tunisia	1 h, 3 h, 6 h, 12 h, 24 h ahead	Point and Interval PV Power Forecasting	nRMSE = 1.33%
Zang et al. (2020b)	2020	CNN-LSTM	China	1 h ahead	Short-term solar radiation prediction	MAE = 37.20 (W/m ²)
Avalos et al. (2020)	2020	CNN, GRU, LSTM, CNN-LSTM, CNN-GRU	Mexico	3-day ahead	Multivariable deep learning models for forecasting in smart grids (including consumption, wind and solar energies)	MAPE = 89.58%
Heo et al. (2021)	2021	Multi-channel CNN	Korea	1-month ahead	Integrate meteorological and geographical features associated with solar energy into the learning process for the prediction of monthly PV outputs	MAPE = 8.639% $R^2 = 0.917$
Netsanet et al. (2022)	2021	VMD-ACO-2NN	China	24 h ahead	Day ahead prediction of power generation	NMSE = 0.0232 $R^2 = 0.9768$
Hossain and Mahmood (2020)	2020	LSTM	USA	6 h, 12 h, 24 h ahead	Multi-step ahead short-term photovoltaic power forecasting	RMSE = (0.94–1.68) (MW) MAE = (0.69–0.82) (MW)
Ma and Zhang (2022)	2022	K-means+EEMD +(MFA-Elman)	China	24 h ahead	Short-term prediction of PV power for the next day	MAE = 0.441 (kW)

in auto-regressive models. TimeGAN accounts for the time dependency in the training data and is also one of the first types of GAN models producing high confidence synthetic time-series data (Plesner, 2021).

Based on the above analysis of the various latest forecasting models and methods, a hybrid short-term PPF model for the PV power stations is proposed in this paper. This model uses a combination of TimeGAN, Soft DTW-based K-medoids clustering, and CNNGRU (Later, it is uniformly named HTSCG). The proposed HTSCG model is compared with GRU (Khatib et al., 2022), CNN-LSTM (Zang et al., 2020b; Alhussein et al., 2020; Bogaerts et al., 2020; Qu et al., 2021; Wang et al., 2019), CNN-GRU (Avalos et al., 2020), K-GASVM (VanDeventer et al., 2019) (Soft DTW-based K-medoids-GASVM) and HKCG (hybrid (Soft DTW-based K-medoids)-(CNN-GRU)) respectively. The main contributions of this paper can be summarized as follows:

- (1) Based on TimeGAN, the effective augmentation of PV power data is realized. The expanded data provide a good foundation for carrying out PV power cluster analysis and improving the final forecasting accuracy.
- (2) An improved K-medoids clustering method is proposed to effectively classify the PV power under different weather conditions. Herein, the normalized Soft DTW-based K-medoids clustering method has been used to enhance the accuracy and rationality of PV power classification.
- (3) A short-term PPF method based on CNN-GRU is proposed. The CNN is applied to intelligently extract the dominant features of PV power, reduce data dimensions, and then make the PPF through the multi-layer GRU network. The CNN-GRU model is an effective alternative to the CNN-LSTM and other hybrid forecasting models in terms of computational complexity as well forecasting accuracy.
- (4) A brand-new forecasting model based on HTSCG is established for short-term PPF. The proposed hybrid models

(“data augmentation-clustering-feature extraction-forecasting”) can always achieve stable multi-step forecasting accuracy under different weather conditions.

The remainder of the paper is organized as follows: Section 2 introduces the structure of the HTSCG model. Section 3 describes the modeling methods proposed in this study. Section 4 describes the experimental data, model evaluation index, and parameter settings. The case study and results analysis are presented in Section 5. Section 6 provides discussions on the advantages and limitations of the proposed model. Section 7 presents the conclusions and suggestions for future work.

2. Design of the TimeGAN-(Soft DTW-based K-medoids)-(CNN-GRU) forecasting model

The structure and data processing flow of the TimeGAN-(Soft DTW-based K-medoids)-(CNN-GRU) model is shown in Fig. 1 with following descriptions:

(1) Data preprocess, including the removal of abnormal data (caused by extreme weather conditions, data transmission errors, etc.), the fill of missing data (caused by equipment failure), and data normalization. since PV generates electricity only during the daytime, here we only extract the diurnal data for the analysis.

(2) Based on the data preprocessing in step (1), the training samples with length L and dimension N (the number of samples during the daytime) are fed into TimeGAN for the data augmentation. The implementation of TimeGAN includes the autoencoding components (embedding and recovery) and the adversarial components (sequence generator and discriminator). The embedding and recovery components link feature and latent space and allow the adversarial network to learn from the underlying temporal data dynamics with lower dimensionality. A more detailed description of the TimeGAN method algorithm is provided in Section 3.1.

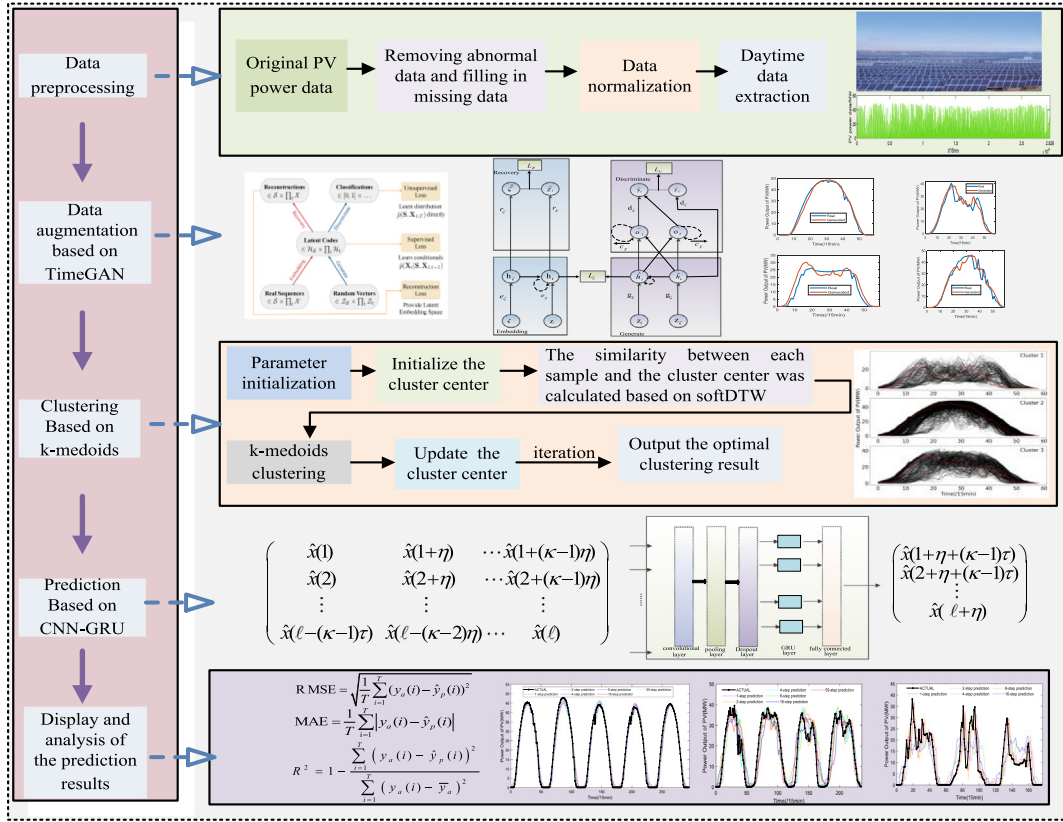


Fig. 1. Structure and data processing scheme of the proposed TimeGAN-(Soft DTW-based K-medoids)-(CNN-GRU) model.

(3) Based on step (2), the augmented data are clustered according to the PV power output characteristics of different weather conditions by using the Soft DTW-based K-medoids clustering method. A detailed description of the implementation of Soft DTW is given in Section 3.2.

(4) The multi-step ahead forecasts are conducted by the CNN-GRU model, where the input and output sample pairs of the training data are generated by the phase space reconstruction method. The network structure of the CNN method, the GRU, and the implementation process of the CNN-GRU hybrid neural networks are presented in Section 3.3.

(5) Presentation of the forecasting results and analysis.

3. Methodologies

3.1. TimeGAN

GAN methods are typically used to generate time series (Chen et al., 2018; Esteban et al., 2017); these methods model $p(\mathbf{X}_{1:T})$ directly by taking the sum of the global GAN loss. To capture the stepwise dependencies in time-series data, the generation model needs to preserve the temporal dynamics of the training data.

TimeGAN combines the advantages of autoregressive models, GAN, and time-series representation learning. As shown in Fig. 2, TimeGAN consists of the autoencoding components and the adversarial components. In the autoencoding components, the embedding network is used to embed the original data into a latent feature space with lower dimensionality, and then recover the space from this feature through the recovery function. The generator is easier to learn the data structure with a lower dimensionality.

TimeGAN contains a global and a local objective (Yoon et al., 2019):

$$\min_{\hat{p}} D((p(\zeta, \mathbf{x}_{1:T})) \| \hat{p}(\zeta, \mathbf{x}_{1:T})) \quad (1)$$

$$\min_{\hat{p}} D((p(\mathbf{x}_t | \zeta, \mathbf{x}_{1:t-1})) \| \hat{p}(\mathbf{x}_t | \zeta, \mathbf{x}_{1:t-1})) \quad (2)$$

where ζ and $\mathbf{x}_{1:T}$ respectively represent the static and dynamic feature vectors of the time series. The objective of Eq. (1) is to use training data to learn a density $\hat{p}(\zeta, \mathbf{x}_{1:T})$ that best approximates $p(\zeta, \mathbf{x}_{1:T})$, and to capture the overall distribution of the training data. Eq. (2) is a local objective and is used to fit the data distribution for each time step; the goal is learning a density $\hat{p}(\mathbf{x}_t | \zeta, \mathbf{x}_{1:t-1})$ that best approximates $p(\mathbf{x}_t | \zeta, \mathbf{x}_{1:t-1})$. Here, the autoregressive decomposition method is used (i.e. $p(\zeta, \mathbf{x}_{1:T}) = p(\zeta) \prod_t p(\mathbf{x}_t | \zeta, \mathbf{x}_{1:t-1})$).

The implementation principle of TimeGAN can be briefly summarized as follows Yoon et al. (2019):

(1) Embedding network between original feature space to latent space

$$\mathbf{h}(\zeta) = \mathbf{e}(\zeta), \mathbf{h}(t) = \mathbf{e}(\mathbf{h}(\xi), \mathbf{h}_{t-1}, \mathbf{x}_t) \quad (3)$$

where $\mathbf{e}$ is the embedding function, which is constructed by a recurrent network. ζ and $\mathbf{x}_T$ represent the static and temporal features of a time series. $\mathbf{h}(\zeta)$ and $\mathbf{h}(t)$ denote the latent representations corresponding to feature spaces $\zeta, \mathbf{x}_T$, which are low-dimensional, dense feature vectors.

(2) Recovery network from latent space to original space

$$\tilde{\zeta} = r_{\zeta}(\mathbf{h}_{\zeta}), \tilde{\mathbf{x}}_t = r_{\mathbf{x}}(\mathbf{h}_t) \quad (4)$$

where r is the recovery network for static and temporal embeddings, constructed by an autoregressive network, and $\tilde{\zeta}$ and $\tilde{\mathbf{x}}_T$ are the accurate reconstructions of $\zeta, \mathbf{x}_T$.

(3) Sequence Generator and Discriminator

The generator of the TimeGAN does not produce results directly in the feature space of the original data; instead, it outputs them into the embedding space, which is critical for improving parameter efficiency.

$$\hat{\mathbf{h}}(\zeta) = g_{\zeta}(\mathbf{z}_{\zeta}), \hat{\mathbf{h}}(t) = g_{\mathbf{x}}(\hat{\mathbf{h}}(\zeta), \hat{\mathbf{h}}_{t-1}, \mathbf{z}_t) \quad (5)$$

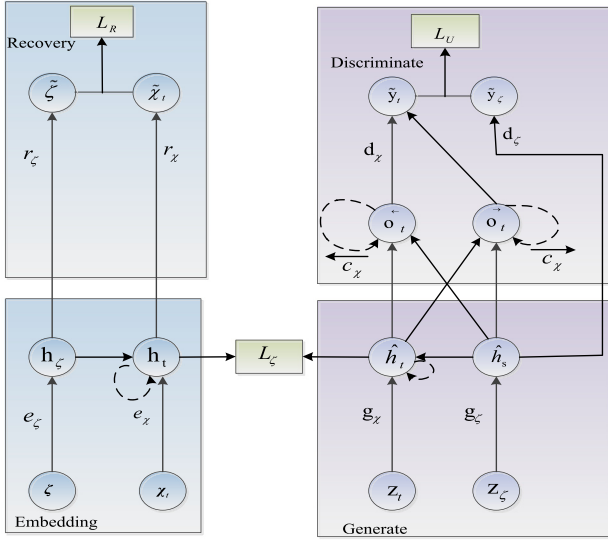


Fig. 2. Network structure of TimeGAN.

where g_s , g_t are the generator networks for static and temporal features, constructed by a recurrent network. z_s , z_t are sequences of uniform random vectors in the range of $[0,1]$.

$$\tilde{y}_s = d_s(\hat{h}_s), \quad \tilde{y}_t = d_t(\hat{h}_t) \quad (6)$$

where $\hat{h}_s$ and $\hat{h}_t$ represent the forward and backward hidden states, respectively; d_s , d_t are the classification functions of output layer; $\tilde{y}_s$ and $\tilde{y}_t$ are the classifications of static and temporal codes.

The loss functions and optimization

(1) The reconstruction loss L_R (Li et al., 2022)

$$L_R = E_{z_s, z_t: T \sim p} \left[\|\zeta - \tilde{\zeta}\|_2 + \sum_t \|\chi_t - \tilde{\chi}_t\|_2 \right] \quad (7)$$

The reconstruction loss L_R was used to test for the accurate reconstruction of the embedding and recovery functions.

(2) The unsupervised loss L_U

The objective function of generator (Li et al., 2022) can be indicated as follows

$$L_U = E_{z_s, z_t: T \sim p} \left[\log y_s + \sum_t \log y_t \right] + E_{z_s, z_t: T \sim \hat{p}} \left[\log(1 - \hat{y}_s) + \sum_t \log(1 - \hat{y}_t) \right] \quad (8)$$

(3) The supervised loss L_z

In order to force the generator to learn the stepwise distribution characteristics in the data, the supervised loss based on the maximum likelihood has been introduced to further control the learning process (Yoon et al., 2019; Li et al., 2022).

$$L_z = E_{z_s, z_t: T \sim p} \left[\sum_t \|\mathbf{h}_t - g_t(\mathbf{h}_s, \mathbf{h}_{t-1}, \mathbf{z}_t)\|_2 \right] \quad (9)$$

TimeGAN follows the following optimizations at the training stage (Li et al., 2022):

$$\min_{\theta_e, \theta_r} (\lambda L_z + L_R) \quad (10)$$

$$\min_{\theta_g} \left(\eta L_z + \max_{\theta_d} L_U \right) \quad (11)$$

where $\theta_e, \theta_r, \theta_g, \theta_d$ represent the parameters of the four networks in TimeGAN. The hyperparameters, λ and η , are used to balance the three loss functions. In this paper, we set $\lambda = 1$, $\eta = 20$.

3.2. Soft DTW-based K-medoids clustering method

3.2.1. K-medoids

K-medoids algorithms belong to the partition-based clustering algorithm, it discards the mean value strategy in K-means and the most centrally located sample was chosen as the clustering center, thus the sensitivity of K-medoids to outliers is reduced. The euclidean distance and DTW distance measurement has been frequently used as a similarity estimation for the K-means and the K-medoids clustering.

3.2.2. Soft DTW

In most cases, two sequences that are overall very similar are not aligned one by one on the x-axis. DTW achieves better alignment by warping the sequences under the time axis.

Let $\mathbf{u} = (u_1, u_2, \dots, u_m)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be the two sequences, the distance between points u_i and v_j can be expressed as $C_{ij} = (u_i - v_j)^2$, and then the local cost matrix $\mathbf{C} \in \mathbb{R}^{m \times n}$ be constructed. The set of each group of adjacent elements in the matrix $\mathbf{C}$ is called “warping path”, defined as $W = w_1, w_2, \dots, w_k, \dots, w_K$, where $\max(m, n) \leq K \leq m + n - 1$. The goal of DTW is to find a path that minimizes the total cumulative distance between the two series $\mathbf{u}$ and $\mathbf{v}$, i.e.,

$$D_{\text{dtw}} = \min_W \sum_{w=1}^K C(w_k). \quad (12)$$

Using the dynamic programming method to calculate the following recurrence, which defines the sum of the distances in the current element and the minimum of the cumulative distances in adjacent elements as the cumulative distance D_{ij} (Cuturi and Blondel, 2017):

$$D_{ij} = |u_i - v_j| + \min(D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}) \quad (13)$$

The DTW distance between time series $\mathbf{u}$ and $\mathbf{v}$ is finally obtained as (Chen et al., 2017):

$$\text{DWT}(\mathbf{u}, \mathbf{v}) = D(m, n) \quad (14)$$

The matrix $\mathbf{A} \in \{0, 1\}^{m \times n}$ is defined as a monotonic alignment matrix, the Frobenius inner product $\langle \mathbf{A}, \mathbf{C} \rangle$ is the sum of costs along the alignment $\mathbf{A}$. The dynamic time warping introduced above can be reformed as follows to solve the minimization problem (Chen et al., 2017):

$$\text{DTW}(\mathbf{C}) = \min_{\mathbf{A} \in \mathcal{A}(m, n)} \langle \mathbf{A}, \mathbf{C} \rangle \quad (15)$$

Soft DTW can differentiate in all of its arguments by the means of taking advantage of a smoothed expression, all possible alignments in $\mathcal{A}(m, n)$ are considered (Cuturi and Blondel, 2017).

$$\min_i^\gamma \{a_i\} = \begin{cases} \min_i \{a_i\} & \gamma = 0 \\ -\gamma \log \sum_i e^{-a_i/\gamma} & \gamma > 0 \end{cases} \quad (16)$$

The parameter γ controls the tuning between smoothness and approximation of the minimum. Substituting the DTW-function (16) into (15), we can obtain the expression of Soft-DTW as follows:

$$\text{DTW}^\gamma(\mathbf{u}, \mathbf{v}) = \min_{\mathbf{A} \in \mathcal{A}(m, n)} \langle \mathbf{A}, \mathbf{C} \rangle = -\gamma \log \sum_{\mathbf{A} \in \mathcal{A}(m, n)} e^{-(\mathbf{A}, \mathbf{C})/\gamma} \quad (17)$$

Soft-DTW is a differentiable alternative to DTW. Soft-DTW can generate negative values due to entropic bias, and is not minimized when the time series are equal. The Soft DTW divergence is

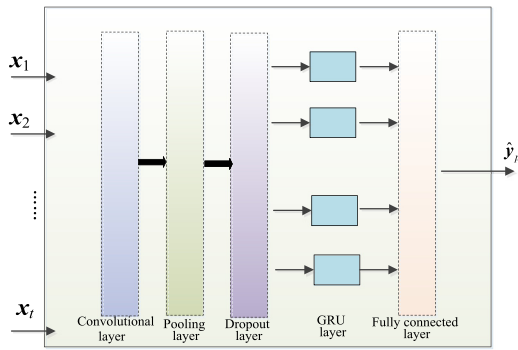


Fig. 3. The CNN-GRU hybrid neural networks.

employed in our experiments to overcome this drawback (Blondel et al., 2021):

$$\text{Div}(\mathbf{u}, \mathbf{v}) = \text{DTW}_\gamma(\mathbf{u}, \mathbf{v}) - \frac{1}{2} \cdot (\text{DTW}_\gamma(\mathbf{u}, \mathbf{u}) + \text{DTW}_\gamma(\mathbf{v}, \mathbf{v})) \quad (18)$$

In our study, the Soft DTW-based K-medoids clustering method is proposed for the PV power time series clustering. Without any prior labels on the acquisition data, time series data of the PV power are divided into several subsequences. Firstly, Soft DTW calculates the similarity among sequences, and then generates the distance matrix. Secondly, the K-medoids method is applied to generate the time series clusters based on the distance matrix that has been computed by the Soft DTW method.

3.3. The CNN-GRU hybrid neural network

CNN-GRU is a hybrid neural network model based on CNN and GRU, a simplified structure of CNN-GRU is presented in Fig. 3. The PV power-related data are taken as the input of the model, and the convolution operation is used to extract the features and capture the data correlation. The number of parameters is then compressed by a pooling operation to further reduce the data dimension. At the same time, the dropout layer is added to randomly select the neurons in the network according to a certain probability to prevent overfitting (Srivastava et al., 2014). Then the GRU layer is used to process the reduced data and capture the time compliance rule among them (Lynn et al., 2019). Finally, the data are reshaped into a one-dimensional sequence by the full connection layer, and then the final output is obtained.

In the proposed framework, we used two CNN layers containing a ReLU activation function and a kernel size of 4, the filter is $2 \times 2 \times 1 \times 8$ and $2 \times 2 \times 8 \times 16$ for the first and second layers, respectively. The features of the pooled layer output are spliced into one-dimensional vectors and input into one dropout layer. Two GRU layers are used to model temporal features and finally, a fully connected layer is used to forecast the future PV power generation.

4. Experiment data and evaluation metrics

4.1. The experimental data

In this paper, the power generation data from 2019 of two PV power stations (They are hereinafter uniformly referred to as station 1 and station 2) located in different regions of Xinjiang, China, were employed to test if the proposed models can be applied for different conditions. The data resolution is 15-min, and the installed capacity of the two PV power stations is 50 MW. The solar energy resources in Xinjiang are very rich, and the number of days with an annual sunshine time of more than 6 h

is in the range of 250–325. Therefore, many PV power stations have been built in this region. The PV power data from January to September were selected as training samples in the following two case studies. The PV power data from October for various weather conditions were selected as test samples to verify the effectiveness and stability of the proposed model. Since the sunrise in Xinjiang is about 07:45 AM and sunset at 21:45 PM, we only extracted the daily data in this period for data augmentation, clustering, and forecasting analysis.

4.2. Model performance evaluation and parameter settings

In this study, the experiment was designed to perform 1-step and multi-step PV power forecasting on different weather conditions. On sunny and cloudy days, the 1-, 2-, 4-, 8-, 16-, and 59-step ahead (one-day) forecasts were conducted; on rainy days, the 15 min to 4 h ahead forecasting was conducted. The five models (i.e., GRU, CNN-LSTM, CNN-GRU, K-GASVM and HKCG), are employed as the reference models to verify the validity of the proposed model. The purpose of selecting them as the reference model was: (1) to verify the forecasting performance of CNGRU model through the GRU and CNNLSTM models; (2) to test the effect of reasonable clustering on improving the forecasting accuracy and verify the forecasting results of CNGRU and HKCG model by comparison; (3) by comparison of the forecasting results with the hybrid forecasting models K-GASVM and HKCG, the forecasting performance of the proposed model will be further proved. In the three single forecasting models, the number of training samples is 15 930 (original data from January to September).

In this study, we use the phase space reconstruction method to build the multi-step PPF model (Li et al., 2021):

$$\hat{\mathbf{y}} = \begin{pmatrix} \hat{x}(1 + \eta + (\kappa - 1)\tau) \\ \hat{x}(2 + \eta + (\kappa - 1)\tau) \\ \vdots \\ \hat{x}(\ell + \eta) \end{pmatrix} = \hat{F} \begin{pmatrix} \hat{x}(1) & \hat{x}(1 + \eta) & \cdots & \hat{x}(1 + (\kappa - 1)\eta) \\ \hat{x}(2) & \hat{x}(2 + \eta) & \cdots & \hat{x}(2 + (\kappa - 1)\eta) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{x}(\ell - (\kappa - 1)\tau) & \hat{x}(\ell - (\kappa - 2)\eta) & \cdots & \hat{x}(\ell) \end{pmatrix} \quad (19)$$

where $\hat{x}(t)$ are the actual PV output power, $F(\cdot)$ represents the CNN-GRU network. κ is the embedding dimension, η is the delay time, and ℓ is the length of the experimental data. The input variable dimensions have been defined as 7 for all models tested; η is also equivalent to the forecasting step size. As shown in Eq. (19), the sampling data taken from historical PV power data at the first 7 time intervals adjacent to the predicted sample point were selected as the input variable. The key parameters of the proposed HTSCG model were determined by the cross-validation method and the insensitive parameters were set to their typical values; in TimeGAN, we normalized the input dataset to the [0,1] range, the embedding network, generator, and discriminator are constructed with 3-layer GRUs, the dimension of the hidden layer is 28, the dimension of the latent space is 4; in Soft DTW-based K-medoids, the number of clustering categories is 3, the smoothing parameter γ is 0.1; in CNN-GRU, the neuron number in both the GRU layer and the full connection layer is 100, the batch size is 32, the dropout rate is 0.3, the learning rate is 0.0001, the maximum number of iterations is 150. The key parameters of the benchmark models are the same as the HTSCG model. For the GASVM model (VanDeventer et al., 2019), since the GA algorithm converges slowly (Katoch et al., 2021), it is difficult to deal with

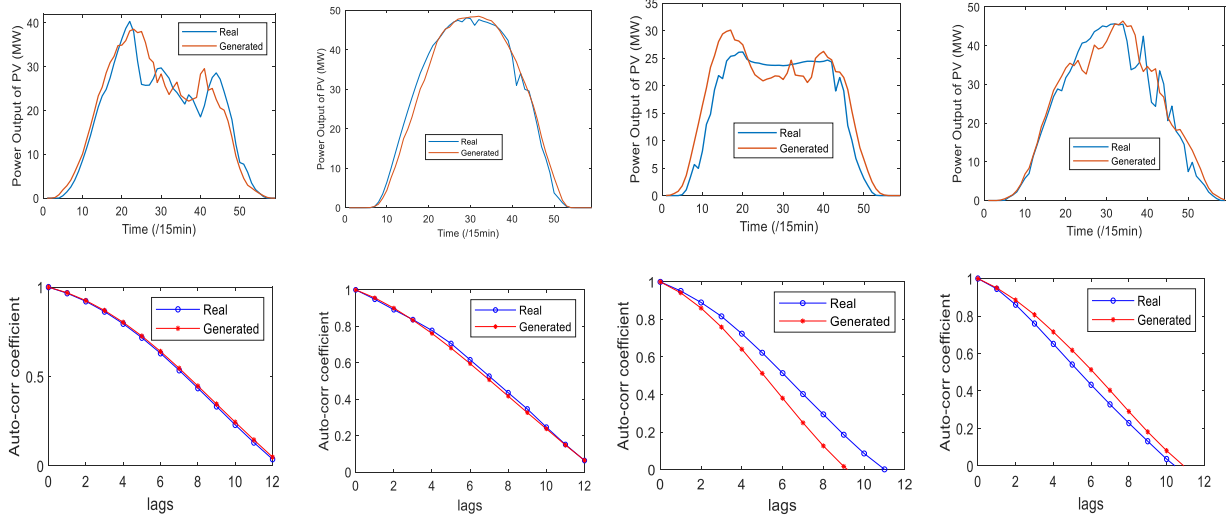


Fig. 4. Real samples versus generated samples (top) and the auto-correlation plots (bottom) for station 1.

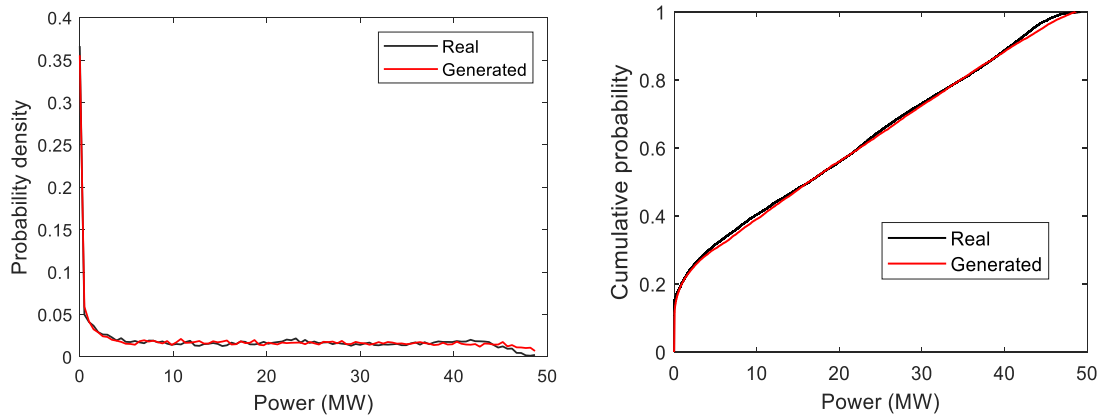


Fig. 5. Probability density (left) and cumulative probability (right) of real samples and generated samples for station 1.

the unclassified data (15 930). In the following case studies, the training data used in GASVM method are based on Soft DTW-based K-medoids. That establishes the K-GASVM method. The optimized parameters of SVM are: kernel function (Gaussian), kernel scale (18), box constraint (175).

To assess the forecasting performance of different methods, the evaluation index of RMSE, mean absolute error (MAE), and the coefficient of determination (R^2) were introduced as follows:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{i=1}^T (y_a(i) - \hat{y}_p(i))^2} \quad (20)$$

$$\text{MAE} = \frac{1}{T} \sum_{i=1}^T |y_a(i) - \hat{y}_p(i)| \quad (21)$$

$$R^2 = 1 - \frac{\sum_{i=1}^T (y_a(i) - \hat{y}_p(i))^2}{\sum_{i=1}^T (y_a(i) - \bar{y}_a)^2}, \bar{y}_a = \frac{1}{T} \sum_{i=1}^T y_a(i) \quad (22)$$

where $y_a(i)$ and $\hat{y}_p(i)$ are the actual and predicted power output at time i , respectively, and T is the number of test sample points, which was 59 in this study.

5. Case study and results analysis

5.1. Data augmentation for PV power

After data clean-up, 270 groups of daytime historical PV power data were obtained from January to September (59 sample points per day). We initially used TimeGANs to generate the PV power time-series with diurnal patterns. The noise vectors $\mathbf{z}_t(270,59)$ and the historical PV power time-series $\mathbf{x}_t$ were fed to the generator and embedding network of TimeGAN, respectively. In this study, we expanded the data of station 1 and station 2 by 4 times and 3 times, respectively. After data augmentation, the number of training samples in station 1 and station 2 are 63 720 and 47 790, respectively.

It can be seen from Figs. 4 and 6 (top) that the generated samples closely resemble real-time samples, and their dynamic fluctuations are well captured. We used the auto-correlation coefficients concerning the time-lag of 0–12, probability density and the cumulative probability to verify the statistical properties of the generated data and the historical data. As shown in Figs. 4 and 6 (bottom), the auto-correlation coefficients of generated samples is very similar to that of the actual samples, which proves

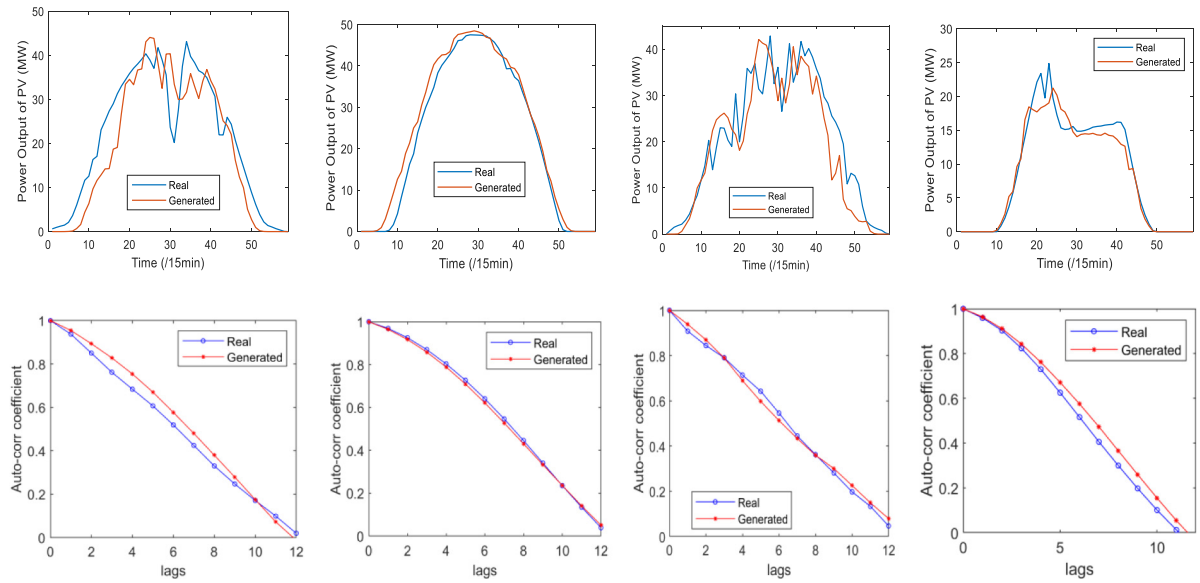


Fig. 6. Real samples versus generated samples (top) and the auto-correlation plots (bottom) for station 2.

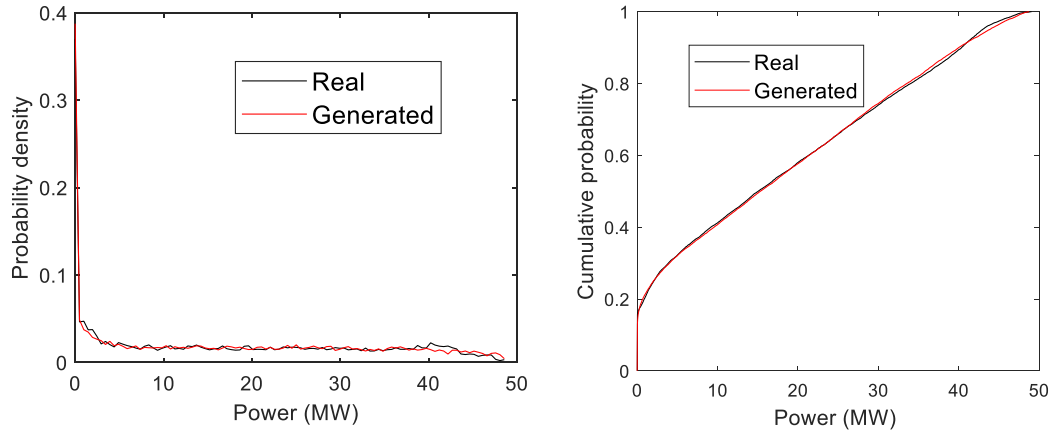


Fig. 7. Probability density (left) and cumulative probability (right) for real and generated samples for station 2.

that the samples generated by TimeGAN can well capture the correlation between the time steps of the original samples. As can be seen from Figs. 5 and 7, the probability density and the cumulative probability of the generated data and the historical data nearly coincide, indicating that TimeGAN not only learned the distribution law of historical data but also learned the shape details of the power curve.

5.2. Clustering analysis for PV power

The augmented data are clustered according to the PV power output characteristics of different weather conditions by using the soft DTW-based K-medoids method. Selection of the category number (k) in K-medoids clustering will directly affect the clustering results. The variations of the sum of squared errors (SSE) and silhouette scores are shown in Fig. 8 for the increasing clustering categories; according to the elbow rule (Chikumbo and Granville, 2019), when the clustering category is 3, the absolute value of second-order derivative for the SSE is the largest and would be suitable as a clustering number. As shown in Fig. 8 for station 1 and station 2, when k is 3, the maximum value of the silhouette score reaches 0.63 and 0.51, respectively. Based on the above analysis, we choose 3 as the number of clustering categories, and the training set is divided into 3 categories. The

numbers of the 3 categories in station 1 were 39 117, 9086, and 15 517 respectively, and the numbers of the 3 categories in station 2 were 18 349, 26 550, and 2891 respectively.

Figs. 9 and 10 show the clustering effect of augmented data and original training data from the two stations, and the red line represents the cluster center. Soft-DTW can always generate smooth centroid curves for different clusters. There are obvious characteristic differences among various categories. These three types of samples represent exactly the power output of the PV station on sunny, cloudy, and rainy days. However, on a comparative basis, the clustering effect of the augmented training data is significantly better due to a larger dataset, and the clustering accuracy is higher (the silhouette scores with respect to the original data of the two stations are 0.54 and 0.47 respectively).

From the clustering results of station 1 and station 2, it is obvious that station 1 has more sunny day samples and station 2 has more cloudy-day samples, but the number of rainy-day samples from both power stations is lower.

5.3. Results analysis

According to Eq. (19), the multi-step ahead forecasting model for PV power was established using the CNN-GRU neural network. Fig. 11 shows the multi-step predicted values and actual value

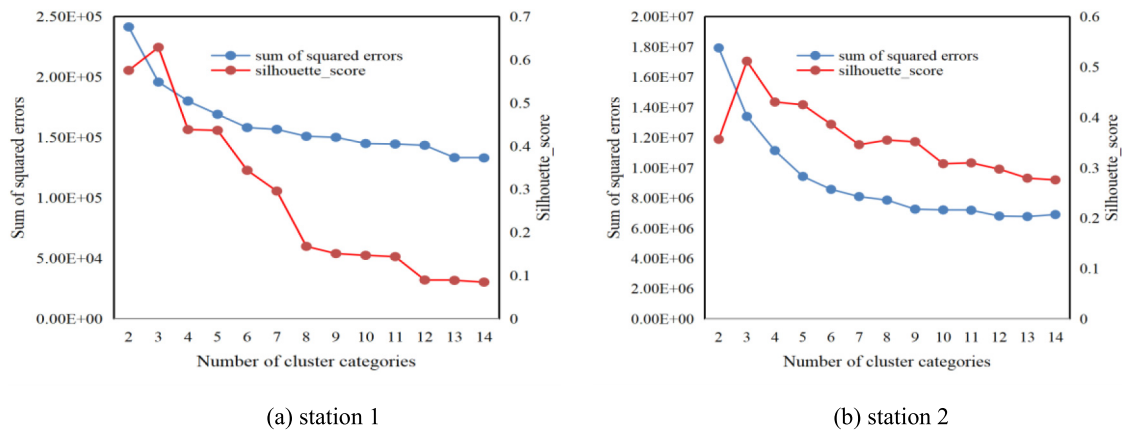


Fig. 8. The sum of squared errors and silhouette score for various clustering categories.

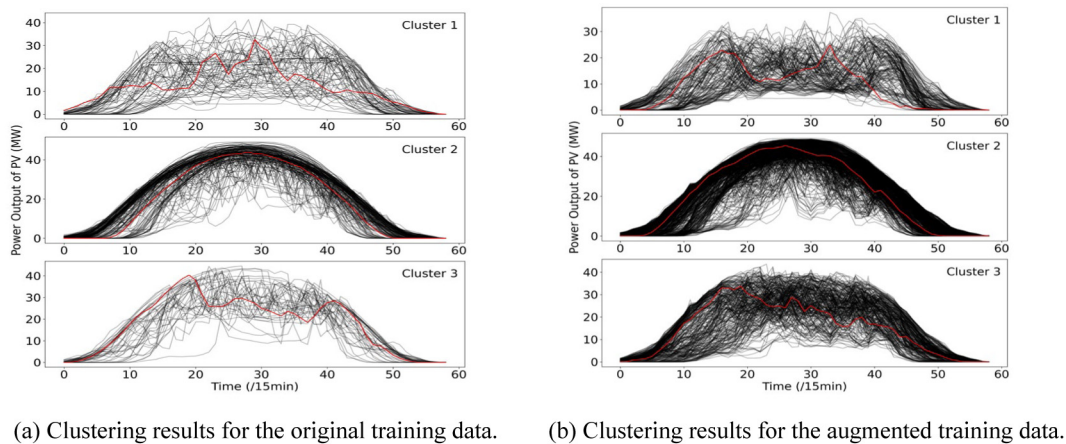


Fig. 9. Clustering results of station 1.

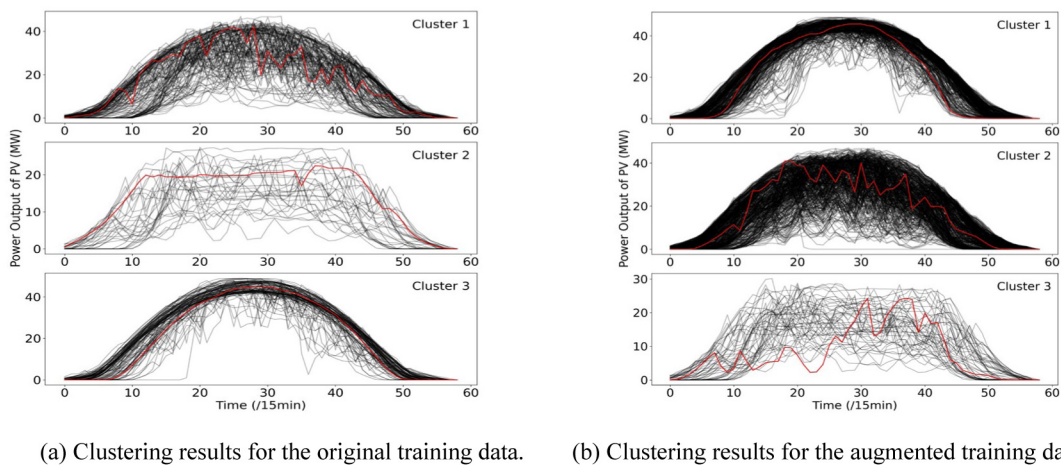


Fig. 10. Clustering results for station 2.

of the proposed model for the two PV power stations. As shown in these graphs, there are great differences in the prediction performance under three different weather types, the forecasting effect of PV power is closely related to the weather conditions. In addition, it can be seen from Fig. 11 that the smaller the forecasting horizon, the higher the forecasting accuracy. This is in line with the autocorrelation law of time series (Massaoudi et al., 2021), because the larger the step size, the weaker the correlation between the input data and the forecasting data.

(1) For sunny days, the proposed model shows good tracking and fitting of the PV power output curve on the two stations; with increasing forecasting steps, small forecasting errors are generated only at individual sample points of power fluctuation. Even if the prediction step is increased to 59 steps, the HTSCG model can still accurately track the shape of the actual power values, the forecasting performance is very stable. (2) For cloudy days, due to the fluctuation of power value, the forecasting performance starts to decline compared with sunny days; in

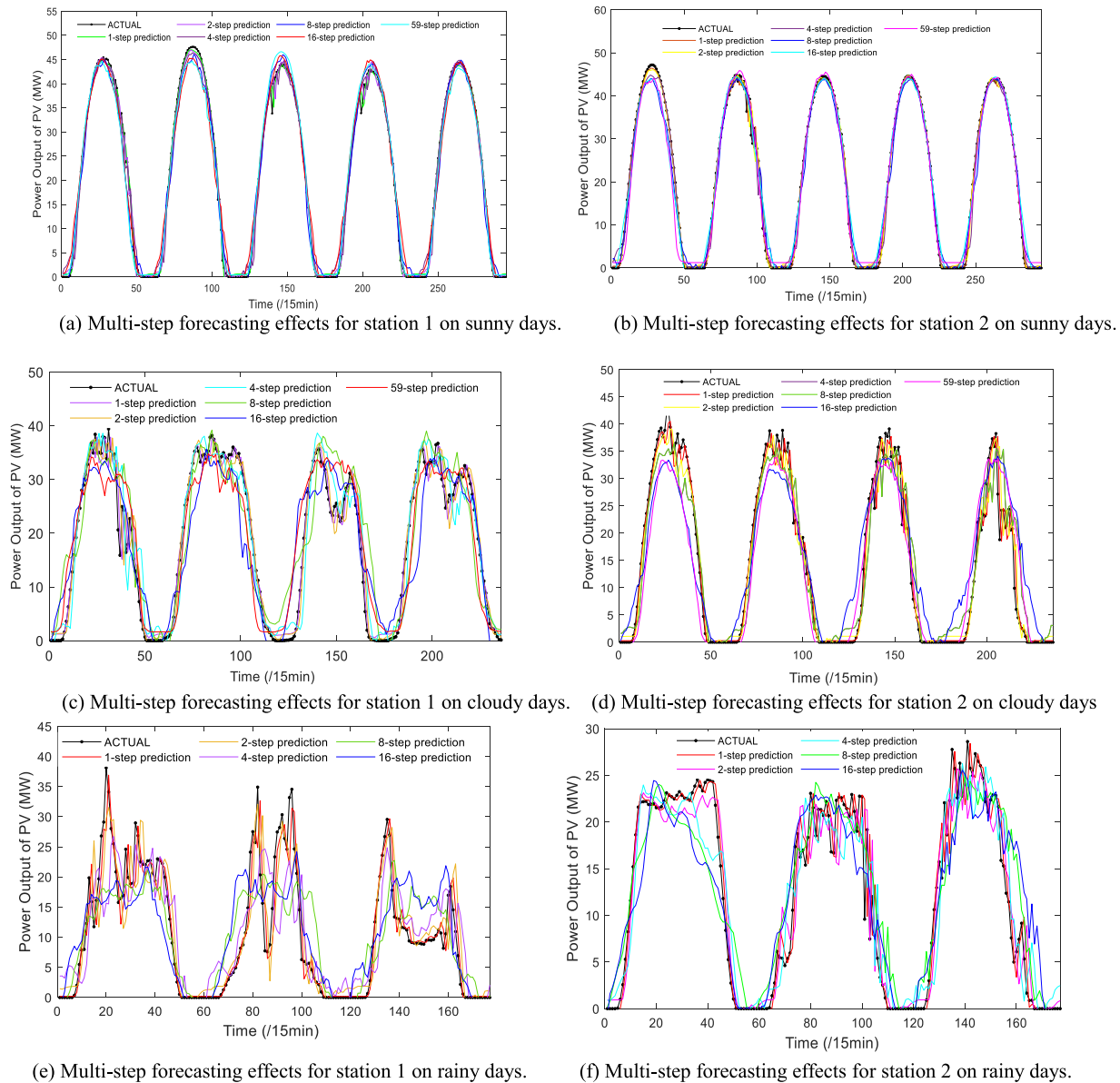


Fig. 11. The multi-step forecasting results of the proposed model versus actual PV power.

multi-step forecasting with more than 4 steps, the corresponding forecasting error is observed within the high fluctuation range of the power curve. (3) The randomness of PV power output on rainy days makes it difficult to execute effective forecasting. It can be seen from Fig. 11(c) that the forecasting skill of HTSCG model dramatically decreases after 8-steps ahead forecasting, so the 59-step ahead forecasting is not considered in the rainy day case study. However, the HTSCG model still exhibits good performance for the 1-step and 2-step forecasting, even at the sudden change of power value, the general shape of the real values is tracked.

The predicted results for the two power stations regarding the sunny days, cloudy days and rainy days are shown in Tables 2–7. The forecasting errors (RMSE, MSE) and the R^2 correlation coefficients with respect to the reference models and the proposed model are presented (the optimized evaluation index for each row is shown in bold). For the two PV stations, it is obvious that the HTSCG model performed the best of the six models under

the three weather conditions. For sunny days, the RMSE values of the proposed model are in the range of (1.003–2.523) MW and (0.927–2.427) MW respectively with respect to the multi-step forecasting of station 1 and station 2; in comparison, the RMSE values of GRU model are in the range of (1.177–4.935) MW and (1.107–3.738) MW respectively. For cloudy days, the MAE value of HTSCG model are in the range of (1.245–4.739) MW and (1.407–4.358) MW; in comparison, the MAE value of CNNLSTM model are in the range of (1.379–5.617) MW and (1.814–5.140) MW. For rainy days, the highest R^2 value of HTSCG model is 0.920 and 0.943, the lowest value is 0.569 and 0.805; in comparison, the highest R^2 value of HKCG model is 0.912 and 0.939, the lowest value is 0.413 and 0.740. These results demonstrate the superior forecasting ability of the proposed model. Due to sufficient sample data of sunny days, HKCG model also shows good performance for the sunny days PV power forecasting.

To intuitively show the improvement for the prediction accuracy from the proposed model, Figs. 12–14 show the contribution

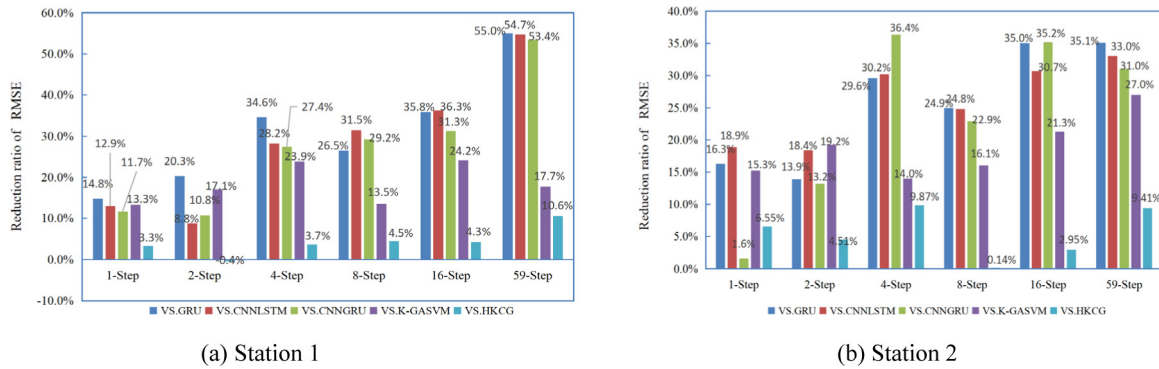


Fig. 12. Reduced percentages (RMSE) of the GRU, CNNLSTM, CNNGRU, K-GASVM and HKCG models compared with the HTSCG models for station 1 and station 2 on sunny days.

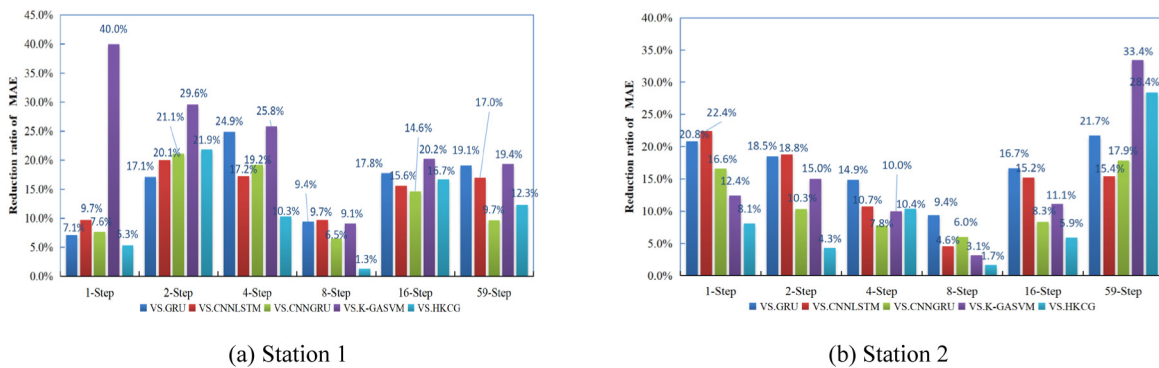


Fig. 13. Reduced percentages (MAE) of the GRU, CNNLSTM, CNNGRU, K-GASVM and HKCG models compared with the HTSCG models for station 1 and station 2 on cloudy days.

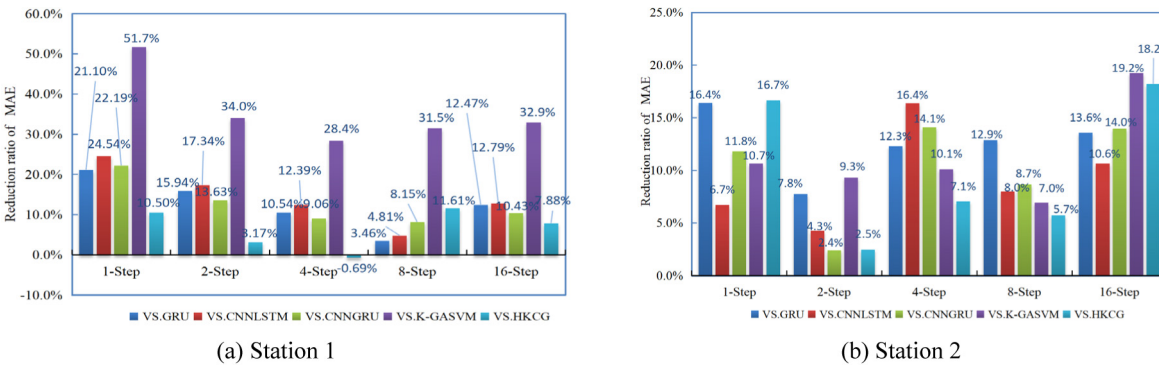


Fig. 14. Reduced percentages (MAE) of the GRU, CNNLSTM, CNNGRU, K-GASVM and HKCG models compared with the HTSCG models for station 1 and station 2 on rainy days.

of the proposed model to the reduced percentage of the forecasting error (RMSE, MAE). For the sunny days, it can be seen from Fig. 12 that the reduced percentages of the RMSE as high as 14.8%–55.0%, 8.8%–54.7%, 10.8%–53.4% and 13.3%–24.2% for station 1 and 13.9%–35.1%, 18.4%–33.0%, 1.6%–36.4% and 14.0%–27.0% for station 2, in comparison with the GRU, CNNLSTM, CNNGRU and K-GASVM models respectively. According to Figs. 13–14, in the one-day ahead forecasting of station 2 on cloudy days, the forecasting error (MAE) of the HTSCG model is 15.4%–33.4% lower than that of the reference models; in the 1-step ahead forecasting for PV power station 1 on rainy days, the forecasting error of the HTSCG model is 10.5–51.7% lower than that of the forecasting models. In contrast to the sunny day forecasting performance, the advantages of the HKCG and K-GASVM models over the three single forecasting models decreased for cloudy

and rainy days, mainly because there are only a few training samples in this class. This lack of data reduces the training effect of CNNGRU and GASVM (In the HKCG model, the training samples from station 1 for sunny, cloudy, and rainy days are 9912, 2242, and 3776 respectively, and 7316, 6785, 1829 training samples from station 2).

Fig. 15 shows the change trend of R^2 correlation coefficients with increasing forecasting steps for the six forecasting models. Apparently, the HTSCG model achieves the highest value in terms of R^2 at each forecasting step. It can be seen from Fig. 15 that the correlation between the predicted value and the observed value of all models shows an overall downward trend with the forecasting horizon enlargement. All methods have shown good forecasting performance in one-step prediction. With the increase of steps, the R^2 value of the three single models has

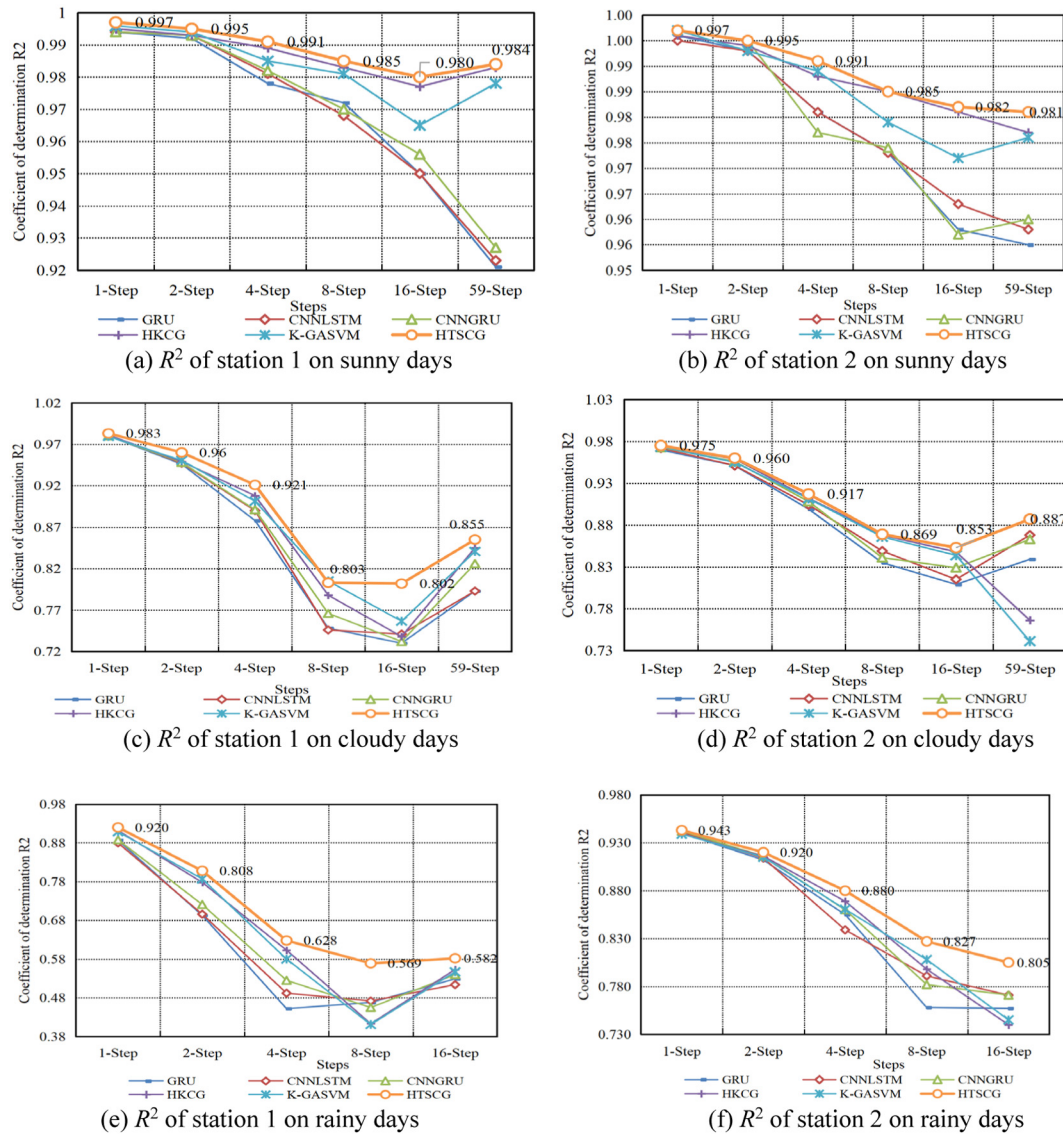


Fig. 15. Performance comparison of the multi-step ahead forecasting of PV power between GRU, CNNLSTM, CNNGRU, K-GASVM, HKCG, and HTSCG models in terms of coefficient of determination (R^2).

declined significantly on three types of samples of two power stations. In contrast, three clustering-based hybrid models maintain a relatively gentle downward trend, which further illustrates the necessity of clustering in PPF. However, it should be pointed out that the forecasting performance of all models for the rainy day of station 1 is not ideal. This is largely related to the random volatility of the predicted sample points. The randomized data of PV power on rainy days make it difficult to obtain effective training samples.

The above results show and compare the overall forecasting results of various models in detail. Herein, the partial steps forecasting effects in regard to every model are displayed in Figs. 16–18 (top), and the corresponding absolute error distribution of six models are displayed (bottom) by box plots. It can be seen from Figs. 16–18 that the fitting ability of HTSCG model to the actual value is significantly better than that of other methods, and the advantages presented by HTSCG model can be easily confirmed through the box plot. For example, it can be clearly seen from Fig. 16(b) and (d) that the predicted value of HTSCG model is closer to the actual value in the power peak period, but the other model like CNNGRU and K-GASVM present overestimation

or underestimation. The forecasting error distribution of HTSCG model is more concentrated (the length of the box is shorter and the lower boundary is closer to 0), and the abnormal forecasting points are relatively less than other models(+).

6. Discussion

The accuracy of PPF is affected by multiple factors such as modeling method, algorithm performance and the quality of input variables, etc. This research proposes a multi-step forecasting model that integrates the benefits of data augmentation, clustering and machine learning method. Results underlined that the HTSCG model possessed better performance than the other five reference models for PPF. It can be seen from Tables 2–7 that the CNNLSTM and CNNGRU models performed better than GRU in the 1-step to one-day ahead forecasting; it reveals that CNN's effective ability of feature extraction is helpful for ANN to improve the forecasting accuracy; GRU is a simplified version of LSTM, which reduces LSTM network complexity while maintaining the same learning performance as LSTM. The combination of CNN and GRU not only improves the learning speed of CNNLSTM, but also

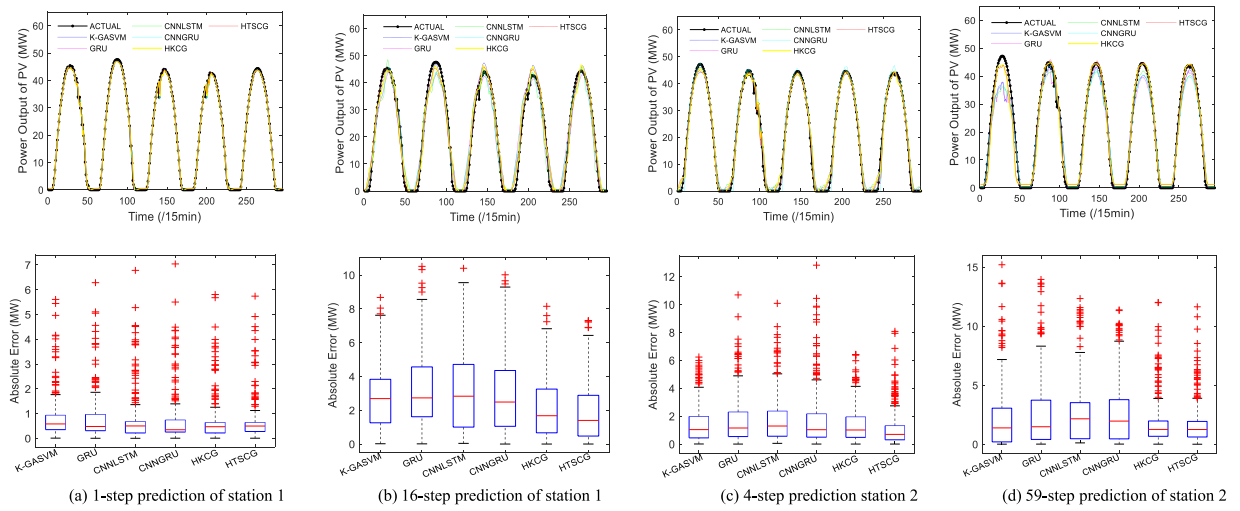


Fig. 16. The prediction results comparison of the HTSCG model vs. the other five reference models on sunny days; top: the forecasting effect, bottom: the corresponding box plot of absolute error distribution.

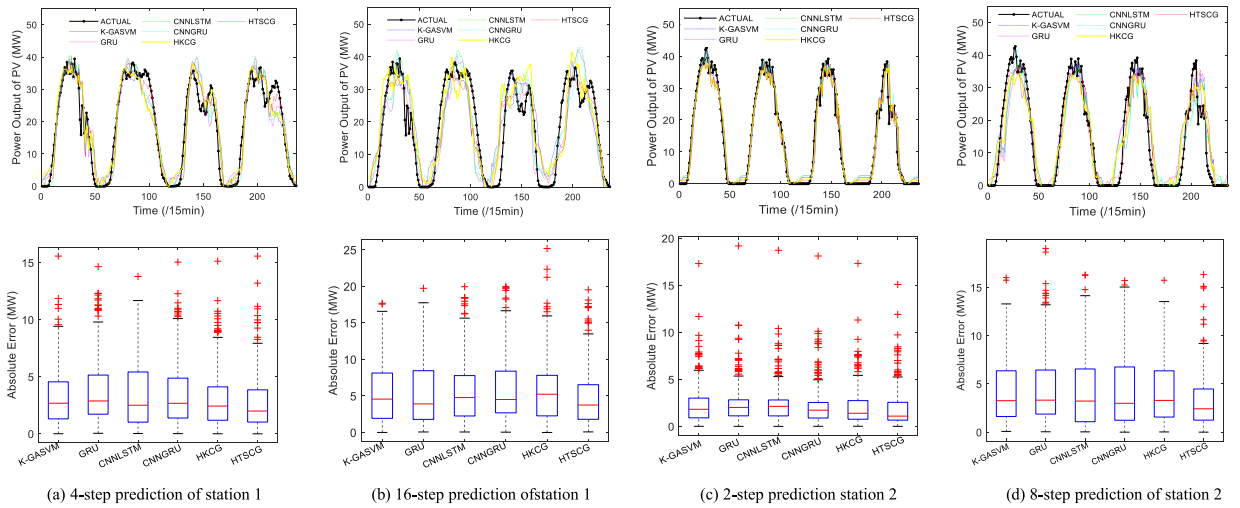


Fig. 17. The prediction results comparison of the HTSCG model vs. the other five reference models on cloudy days; top: the forecasting effect, bottom: the corresponding box plot of absolute error distribution.

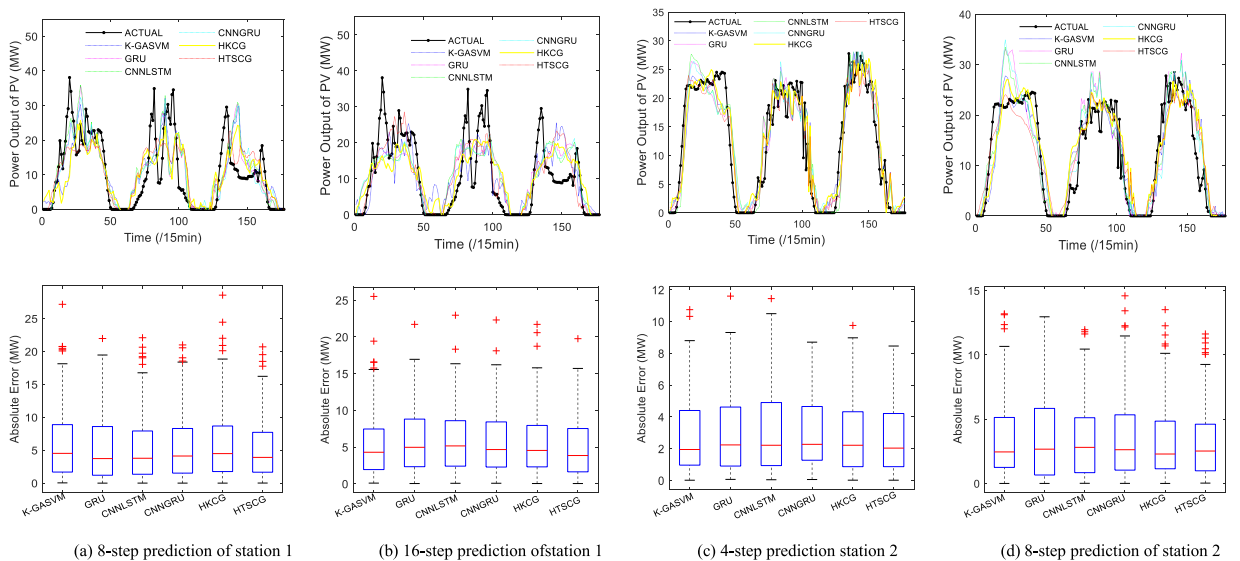


Fig. 18. The prediction results comparison of the HTSCG model vs. the other five reference models on rainy days; top: the forecasting effect, bottom: the corresponding box plot of absolute error distribution.

Table 2
Performance evaluations of different PPF models with respect to power station 1 (sunny days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	1.177	1.152	1.136	1.156	1.037	1.003
	MAE (MW)	0.776	0.698	0.673	0.808	0.633	0.643
	R^2	0.994	0.994	0.994	0.996	0.995	0.997
2-Step (30 min)	RMSE (MW)	1.497	1.308	1.337	1.438	1.188	1.193
	MAE (MW)	1.062	0.819	0.815	0.999	0.790	0.792
	R^2	0.992	0.993	0.993	0.994	0.993	0.995
4-Step (1 h)	RMSE (MW)	2.606	2.376	2.347	2.240	1.770	1.705
	MAE (MW)	1.951	1.857	1.762	1.638	1.215	1.230
	R^2	0.978	0.981	0.982	0.985	0.989	0.991
8-Step (2 h)	RMSE (MW)	2.907	3.118	3.019	2.472	2.238	2.137
	MAE (MW)	2.149	2.292	2.336	1.875	1.641	1.585
	R^2	0.972	0.968	0.970	0.981	0.983	0.985
16-Step (4 h)	RMSE (MW)	3.930	3.960	3.670	3.327	2.636	2.523
	MAE (MW)	3.237	3.156	2.937	2.774	2.051	1.880
	R^2	0.950	0.950	0.956	0.965	0.977	0.980
59-Step (one-day ahead)	RMSE (MW)	4.935	4.901	4.760	2.698	2.481	2.219
	MAE (MW)	3.375	3.664	3.556	2.155	1.699	1.601
	R^2	0.921	0.923	0.927	0.978	0.983	0.984

Table 3
Performance evaluations of different PPF models with respect to power station 1 (cloudy days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	1.963	1.888	1.927	2.490	1.900	1.781
	MAE (MW)	1.340	1.379	1.348	2.074	1.315	1.245
	R^2	0.980	0.981	0.980	0.979	0.981	0.983
2-Step (30 min)	RMSE (MW)	3.203	3.194	3.160	3.304	3.130	2.765
	MAE (MW)	2.314	2.399	2.432	2.724	2.455	1.918
	R^2	0.946	0.947	0.948	0.951	0.949	0.960
4-Step (1 h)	RMSE (MW)	4.846	4.604	4.579	4.515	4.197	3.887
	MAE (MW)	3.794	3.441	3.524	3.839	3.175	2.848
	R^2	0.878	0.890	0.891	0.901	0.908	0.921
8-Step (2 h)	RMSE (MW)	6.963	6.989	6.715	6.294	6.387	6.154
	MAE (MW)	5.282	5.300	5.117	5.266	4.849	4.786
	R^2	0.748	0.746	0.766	0.805	0.788	0.803
16-Step (4 h)	RMSE (MW)	7.212	7.068	7.185	6.978	7.109	6.173
	MAE (MW)	5.764	5.617	5.552	5.941	5.688	4.739
	R^2	0.730	0.741	0.732	0.757	0.738	0.802
59-Step (one-day ahead)	RMSE (MW)	6.306	6.309	5.786	5.811	5.461	5.282
	MAE (MW)	5.026	4.900	4.501	5.045	4.638	4.066
	R^2	0.793	0.793	0.826	0.841	0.845	0.855

can obtain higher forecasting accuracy (Avalos et al., 2020; Sajjad et al., 2020).

In addition to the superiority of the CNNGRU over the comparative CNNLSTM model and GRU, the present results of HKCG and K-GASVM (VanDeventer et al., 2019) model demonstrated the suitability of clustering method, where the performance metrics (Tables 2–7) for the clustering-based models were noticeably better than three standalone models on majority of forecasting steps, but it does not maintain a stable performance for all the forecasting horizons. this reveals that clustering alone is not enough when the training data is insufficient. Further analysis the forecasting results of K-GASVM and HKCG, we can see that the HKCG outperforms the K-GASVM, it indirectly proves the forecasting performance of CNNGRU model.

In addition to integrating CNNGRU with Soft DTW-based K-medoids clustering, a further improvement on forecasting accuracy was attained by incorporation of data augmentation that resulted in the novel TimeGAN- (Soft DTW-based K-medoids)- (CNN-GRU)) forecasting model. GAN and its variants (CGAN, RCGAN, FeTGAN) has been used in time series data generation (Esteban et al., 2017; Plesner, 2021; Ramponi et al.,

2018); yet in this study, TimeGAN has been implemented for data augmentation of PV power time series. Figs. 4 and 6 shows that TimeGAN not only generate realistic PV power profiles with full diversity of behaviors but also capture the stepwise dependencies present in the training data. The effective augmenting of the training data by TimeGAN lead to a improved overall forecasting performance of the HTSCG method as compared to HKCG . It should be noted that the proportion of forecasting accuracy improvement contributed by HTSCG model is reduced on rainy days of station 1, which is closely related to the quality of the generated samples. With the randomness of the original data increasing, a deviation appearance between the generated data and the actual data, which affects the forecasting accuracy; this is a limitation of this study, which makes it difficult for the HTSCG model implement longer time horizon forecasting for the rainy day.

The proposed model in this paper is a brand-new PV power forecasting method, the modeling method is obviously different from the hybrid models in Ref. De Giorgi et al. (2016), Zhang et al. (2021), Netsanet et al. (2022) and Ma and Zhang (2022). An important finding of this paper is that the proposed hybrid

Table 4

Performance evaluations of different PPF models with respect to power station 1 (rainy days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	3.428	3.536	3.421	3.879	3.025	2.896
	MAE (MW)	2.085	2.180	2.114	3.404	1.838	1.645
	R ²	0.887	0.880	0.888	0.909	0.912	0.920
2-Step (30 min)	RMSE (MW)	5.669	5.642	5.407	5.277	4.815	4.482
	MAE (MW)	3.702	3.765	3.603	4.718	3.214	3.112
	R ²	0.693	0.696	0.721	0.787	0.778	0.808
4-Step (1 h)	RMSE (MW)	7.575	7.294	7.050	7.056	6.444	6.237
	MAE (MW)	5.074	5.181	4.991	6.337	4.508	4.539
	R ²	0.452	0.492	0.525	0.580	0.603	0.628
8-Step (2 h)	RMSE (MW)	7.466	7.435	7.553	8.376	7.842	6.621
	MAE (MW)	5.347	5.423	5.620	7.538	5.840	5.162
	R ²	0.468	0.472	0.455	0.411	0.413	0.569
16-Step (4 h)	RMSE (MW)	7.024	7.137	6.938	7.754	6.842	6.595
	MAE (MW)	5.742	5.763	5.611	7.495	5.456	5.026
	R ²	0.529	0.514	0.540	0.547	0.553	0.582

Table 5

Performance evaluations of different PPF models with respect to power station 2 (sunny days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	1.107	1.143	0.942	1.094	0.992	0.927
	MAE (MW)	0.739	0.817	0.539	0.780	0.625	0.499
	R ²	0.996	0.995	0.997	0.997	0.996	0.997
2-Step (30 min)	RMSE (MW)	1.403	1.480	1.392	1.495	1.265	1.208
	MAE (MW)	0.953	1.120	0.991	1.070	0.868	0.748
	R ²	0.993	0.993	0.995	0.993	0.994	0.995
4-Step (1 h)	RMSE (MW)	2.412	2.433	2.670	1.976	1.885	1.699
	MAE (MW)	1.710	1.764	1.751	1.447	1.394	1.121
	R ²	0.981	0.981	0.977	0.989	0.988	0.991
8-Step (2 h)	RMSE (MW)	2.894	2.889	2.818	2.588	2.175	2.172
	MAE (MW)	2.072	2.184	2.012	2.067	1.698	1.611
	R ²	0.973	0.973	0.974	0.979	0.985	0.985
16-Step (4 h)	RMSE (MW)	3.640	3.415	3.650	3.005	2.438	2.366
	MAE (MW)	2.698	2.659	2.801	2.420	1.754	1.731
	R ²	0.958	0.963	0.957	0.972	0.981	0.982
59-Step (one-day ahead)	RMSE (MW)	3.738	3.625	3.518	3.324	2.679	2.427
	MAE (MW)	2.493	2.573	2.536	2.170	1.823	1.672
	R ²	0.955	0.958	0.960	0.976	0.977	0.981

Table 6

Performance evaluations of different PPF models with respect to power station 2 (cloudy days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	2.538	2.464	2.404	2.446	2.352	2.333
	MAE (MW)	1.777	1.814	1.687	1.607	1.531	1.407
	R ²	0.970	0.972	0.973	0.974	0.974	0.975
2-Step (30 min)	RMSE (MW)	3.260	3.260	3.091	3.213	2.975	2.950
	MAE (MW)	2.438	2.448	2.216	2.338	2.077	1.987
	R ²	0.951	0.951	0.956	0.955	0.959	0.960
4-Step (1 h)	RMSE (MW)	4.710	4.598	4.502	4.485	4.402	4.256
	MAE (MW)	3.547	3.382	3.275	3.354	3.368	3.019
	R ²	0.899	0.904	0.908	0.911	0.912	0.917
8-Step (2 h)	RMSE (MW)	6.028	5.767	5.926	5.466	5.376	5.286
	MAE (MW)	4.593	4.362	4.429	4.297	4.233	4.162
	R ²	0.835	0.849	0.841	0.866	0.868	0.869
16-Step (4 h)	RMSE (MW)	6.495	6.389	6.135	5.929	5.780	5.499
	MAE (MW)	5.230	5.140	4.753	4.902	4.631	4.358
	R ²	0.809	0.815	0.829	0.844	0.848	0.853
59-Step (one-day ahead)	RMSE (MW)	5.952	5.395	5.501	7.701	7.180	5.099
	MAE (MW)	4.685	4.336	4.466	5.511	5.125	3.668
	R ²	0.839	0.868	0.863	0.741	0.766	0.887

Table 7
Performance evaluations of different PPF models with respect to power station 2 (rainy days).

Forecasting horizon	Evaluation index	Forecasting models					
		GRU	CNNLSTM	CNNGRU	K-GASVM	HKCG	HTSCG
1-Step (15 min)	RMSE (MW)	2.436	2.419	2.395	2.483	2.458	2.384
	MAE (MW)	1.657	1.485	1.571	1.550	1.662	1.385
	R ²	0.940	0.941	0.942	0.939	0.939	0.943
2-Step (30 min)	RMSE (MW)	2.961	2.927	2.887	2.931	2.896	2.812
	MAE (MW)	2.232	2.151	2.110	2.271	2.112	2.059
	R ²	0.912	0.914	0.916	0.915	0.916	0.920
4-Step (1 h)	RMSE (MW)	3.798	4.009	3.737	3.741	3.614	3.353
	MAE (MW)	2.940	3.085	3.002	2.870	2.775	2.579
	R ²	0.855	0.839	0.860	0.861	0.869	0.880
8-Step (2 h)	RMSE (MW)	4.914	4.562	4.660	4.448	4.493	4.048
	MAE (MW)	3.654	3.461	3.486	3.421	3.376	3.183
	R ²	0.758	0.791	0.782	0.808	0.798	0.827
16-Step (4 h)	RMSE (MW)	4.923	4.777	4.982	5.115	5.094	4.337
	MAE (MW)	3.817	3.691	3.834	4.083	4.031	3.298
	R ²	0.757	0.771	0.771	0.745	0.740	0.805

forecasting method has more obvious advantages in larger step forecasting. However, other research methods are easier to improve the forecasting accuracy in a smaller step forecasting. For example, the improvement Ref. [Zhang et al. \(2021\)](#) is mainly in the 1-step and 2-step forecasting, and the improvement of Ref. [Massaoudi et al. \(2021\)](#) is mainly in 1-h ahead and 2-h ahead forecasting. Combined with the advantages of the HTSCG model in the 24-h ahead forecasting of sunny days, it reveals that this method has the ability to be popularized to other time series forecasting fields with larger step size and strong regularity, such as day ahead load forecasting, etc. Finally, it should be pointed out that the combination of multiple methods virtually increases the complexity of forecasting model, which is a common problem faced by most hybrid forecasting methods.

7. Conclusion and future work

A novel hybrid model for short-term PV power output forecasting is proposed and presented in this paper. This hybrid model uses a combination of TimeGAN, improved K-medoids clustering algorithms, and a CNN-GRU neural network, namely HTSCG. The performance of the proposed model is investigated in the multi-step forecasting of two PV power stations under three weather conditions. The results show that:

(1) PV power output is significantly affected by meteorological factors, and apparent differences in power value, power curve trend, and power fluctuation frequency can be observed under different weather conditions. The comparison of the forecasting results of HKCG and CNNGRU shows that the improved K-medoids clustering method improves the modeling efficiency and forecasting accuracy.

(2) Through initial technical processing and augmentation of the original data, a sufficiently large number of training sample data that reproduce the power fluctuations of the predicted day can be obtained. The forecasting results from sunny days reveal that the HKCG forecasting model is also very effective when the training dataset is relatively large, but the forecasting results on cloudy days of power station 1 and the forecasting results on rainy days of power station 2 show that when the training sample number is insufficient, the forecasting accuracy of the HKCG model will decline significantly. The advantage of the HTSCG model over the HKCG model is that the data expansion improves the accuracy of PPF on cloudy and rainy days.

(3) The experimental results show that the forecasting performance of the HTSCG model is very stable, and the forecasting accuracy is significantly higher than that of other reference models. The HTSCG model is more practical in new PV power stations

or PV power stations with insufficient historical operation data. The multi-step forecasting results of HTSCG model show that the data processing strategy, from data augmentation to clustering, is an effective time series modeling method, which could be applied to wind power and load forecasting.

The focus of future research will be on the TimeGAN algorithm to obtain better quality PV power generation samples for cloudy and rainy days. Future work will consider to add the numerical weather prediction data to input variables to achieve better forecasting accuracy for cloudy and rainy days.

CRedit authorship contribution statement

Qing Li: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Xinyan Zhang:** Validation, Formal analysis, Visualization, Software. **Tianjiao Ma:** Validation, Experiment analysis, Visualization. **Dagui Liu:** Resources, Writing – review & editing, Supervision, Data curation. **Heng Wang:** Writing – review & editing. **Wei Hu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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